SOFTWARE REQUIREMENTS
SPECIFICATION

Smart Queue Management System (SQMS)

Milestone 2 — Requirements Engineering and System Analysis
Team 6 (T6) | Repository: https://github.com/namahorojanvier1986-blip/secd-2026-team6-sqms
Version: Final submission draft | Date: 13 November 2026

Prepared from the Team 6 M1 problem-discovery evidence and the M2 requirements baseline.

Document Control

Purpose: define the functional and non-functional requirements, use cases, user stories, priorities and traceability needed to guide implementation and testing of SQMS.

Field	Value
Project	Smart Queue Management System (SQMS)
Team	Team 6 (T6)
Milestone	M2 — Requirements
Repository	https://github.com/namahorojanvier1986-blip/secd-2026-team6-sqms
Team roles	Jeavis — Project Manager; Janvier Namahoro — UX/UI; Eraste Hakizimana — QA (confirm current allocation with team).
Version status	Prepared for team review and submission. Validation details and FR-21 wording should be checked against the team's signed-off notes.

Revision notes

This version carries forward FR-01 to FR-20, NFR-01 to NFR-14, UC-01 to UC-09 and US-01 to US-14 from the recovered M2 SRS. It incorporates the stated corrections: US-08 acceptance criteria separated into AC-1 and AC-2; US-10 raised from 2 to 5 points; FR-20 status keywords extended to four languages; FR-21 added; and MoSCoW total recalculated to 49 points.

Important accuracy note: the original recovered SRS was 9 pages. The team's later statement refers to a revised 15-page version, but the detailed V1/V2 response notes and exact FR-21 wording were not present in the recovered source files. This document therefore identifies those two items for team confirmation rather than inventing participant quotations or claiming independent validation.

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) states what SQMS must do and the quality constraints it must meet. It is intended for developers, testers, designers, reviewers and stakeholders. The structure is inspired by ISO/IEC/IEEE 29148:2018 [R1].

1.2 Scope

In scope: single-branch ticket requests through a tablet-friendly web page; sequential queue numbers; estimated waiting time; SMS confirmation, turn-near and called-to-counter notifications; priority handling; teller queue operations; manager dashboard and CSV export; English and Kinyarwanda customer screens; staff roles; service and counter configuration; and audit records.

Out of scope: core banking integration, appointment booking, online payments, native mobile application, kiosk/ticket-printer hardware manufacture, biometric or national-ID verification, multi-branch head-office analytics, live commercial SMS contracts and production deployment.

1.3 Definitions

Ticket: a queue number issued to one customer for one service. Priority customer: an elderly person, pregnant woman or person with a disability under the institution's policy. Estimated wait (ETW): customers ahead multiplied by recent average service time. No-show: a customer who does not reach the counter within the grace period after being called. FR/NFR: functional/non-functional requirement.

1.4 References and governing assumptions

The M1 evidence base comprises five anonymised interviews conducted in Rwamagana on 5 November 2026 and the published/industry sources [S1]–[S9]. Personal data handling must be reviewed against Rwanda Law No. 058/2021 [R2]. Accessibility targets follow WCAG 2.2 [R3].

2. Overall Description

2.1 Product perspective

SQMS is a responsive web application for one financial-institution branch. Customers use a branch tablet; tellers and managers use desktop or laptop browsers. The back end exposes a REST API and persists queue records in PostgreSQL. SMS is integrated through a gateway sandbox during the course.

2.2 User classes and needs

Actor	Main needs	M1 evidence
Customer	Know queue position and wait; leave and return; trust fair order.	P1, P3
Priority customer	Receive priority service without repeatedly asking staff.	P2
Teller	See waiting list, call next, mark served/no-show and recall.	P4
Branch manager	Monitor waiting, service volume and peak periods.	P5
Administrator	Configure services, counters, rules and thresholds.	Operational role
SMS gateway	Deliver queue and status messages.	External service

2.3 Operating environment

Customer tablet: approximately 10-inch Android device with a current browser. Staff: desktop/laptop using the latest two versions of Chrome or Edge. Backend: Node.js 20 with Express; PostgreSQL database; HTML, CSS and JavaScript responsive client; Jest automated tests. SMS gateway is simulated/sandboxed.

2.4 Constraints and assumptions

The team has limited build/test sessions; Must stories are the core scope. No implementation begins before design approval at the end of Session 3. Branch power and network are assumed stable during opening hours. Institutions provide their own service categories and priority policy. Customers without phones are supported by a floor guide. Device/connectivity details not captured in M1 remain assumptions to validate.

3. Functional Requirements (FR-01 to FR-11)

FR-01 — Must | Source: PP1, PP3

The system shall let a customer request a ticket by selecting a service type and entering a mobile phone number on the branch device.

FR-02 — Must | Source: Privacy [R2]

The system shall display a consent notice explaining that the phone number is used only for queue SMS, and require acceptance before issuing a ticket.

FR-03 — Must | Source: PP3

The system shall generate a unique sequential queue number per branch, service and day automatically.

FR-04 — Must | Source: PP1

The system shall send an SMS confirming the queue number, service and estimated wait within seconds of ticket issue.

FR-05 — Must | Source: PP1

The system shall calculate estimated wait from the number of customers ahead and the recent average service time, and update it as the queue moves.

FR-06 — Must | Source: PP1

The system shall send an SMS when a configurable number of customers (default 3) remain ahead of the customer.

FR-07 — Should | Source: PP1

The system shall send an SMS naming the counter when a customer is called.

FR-08 — Must | Source: PP2

The system shall let an eligible customer choose a priority category (elderly, pregnant, person with disability) and place priority tickets ahead of standard tickets according to institution policy.

FR-09 — Must | Source: PP3, PP4

The system shall let a teller view the waiting list for their service and call the next ticket (priority first, then first-come-first-served).

FR-10 — Must | Source: PP3, PP4

The system shall let a teller mark a ticket as served, recall a called ticket, or mark it as no-show after the grace period (default 5 minutes).

FR-11 — Must | Source: PP1

The system shall keep a customer's place while they are away until called, and let a no-show customer rejoin at the next available position once.

3. Functional Requirements (FR-12 to FR-21)

FR-12 — Must | Source: PP5

The system shall record timestamps for ticket created, called, service started and completed.

FR-13 — Must | Source: PP5

The system shall give a manager a live dashboard showing customers waiting, average wait, customers served and busiest hours for the current day.

FR-14 — Should | Source: PP5

The system shall let a manager export queue statistics by day and hour as CSV.

FR-15 — Should | Source: PP1, PP2

The system shall present customer-facing kiosk screens and SMS texts in Kinyarwanda and English, selectable by the customer.

FR-16 — Must | Source: PP4, PP5

The system shall require staff login with roles (teller, manager, administrator) and restrict functions by role.

FR-17 — Should | Source: PP2, PP5

The system shall let an administrator configure services, counters, priority rules and notification thresholds.

FR-18 — Should | Source: PP2

The system shall let the floor guide or teller issue a ticket for a customer without a phone number; the ticket number is shown on the hall display.

FR-19 — Could | Source: PP3

The system shall show currently served numbers and counters on a hall display screen.

FR-20 — Could | Source: PP1

The system shall let a customer reply to an SMS keyword to receive the current ticket status; recognised keywords are localised in English, Kinyarwanda, French and Swahili.

FR-21 — Should | Source: PP4, PP5; privacy/accountability

The system shall maintain an audit trail of staff actions that change ticket status or queue configuration, recording the actor, action, ticket or configuration reference, and timestamp.

FR-20 language keyword examples

The status-reply function must recognise a keyword in the language selected by the customer. Example terms for team review: STATUS (English), IMIMERERE (Kinyarwanda), STATUT (French), and HALI (Swahili). Confirm the exact Swahili wording with a fluent reviewer before release.

FR-21 confirmation

FR-21 is drafted here as an audit-trail requirement to make staff actions traceable. The team must compare this wording with the exact approved FR-21 from its validation/change log before treating the requirement as final.

4. Non-Functional Requirements

NFR-01 — Performance

A ticket is issued and shown on the kiosk within 3 seconds for 95% of requests with 50 customers using the system at once.

NFR-02 — Performance

An SMS is handed to the gateway within 10 seconds of the triggering event for 95% of messages.

NFR-03 — Performance

The teller screen and manager dashboard refresh within 2 seconds of a queue change.

NFR-04 — Availability

The system is available at least 99% of branch opening hours (08:00–17:00) per month.

NFR-05 — Privacy

Phone numbers are collected only with consent, shown masked to staff (last 3 digits), and deleted or anonymised after 30 days, subject to legal review.

NFR-06 — Security

All traffic uses TLS 1.2 or higher; staff passwords are stored hashed; sessions expire after 15 minutes of inactivity; access is role-based.

NFR-07 — Usability

At least 80% of first-time customers complete ticket registration in under 60 seconds without help (tested in M5 with at least 3 users).

NFR-08 — Accessibility

Text contrast is at least 4.5:1, touch targets are at least 44 px, every input has a label; web screens target WCAG 2.2 AA.

NFR-09 — Localisation

All customer-facing text is available in Kinyarwanda and English; each SMS fits one 160-character segment. Status keywords support four languages under FR-20.

NFR-10 — Accuracy

Estimated wait is within 10 minutes of actual wait for at least 80% of tickets in simulated test runs.

NFR-11 — Scalability

One branch supports at least 500 tickets per day and 5 counters without degrading NFR-01.

NFR-12 — Data integrity

No two tickets share the same number for the same branch, service and day, including under concurrent requests.

NFR-13 — Compatibility

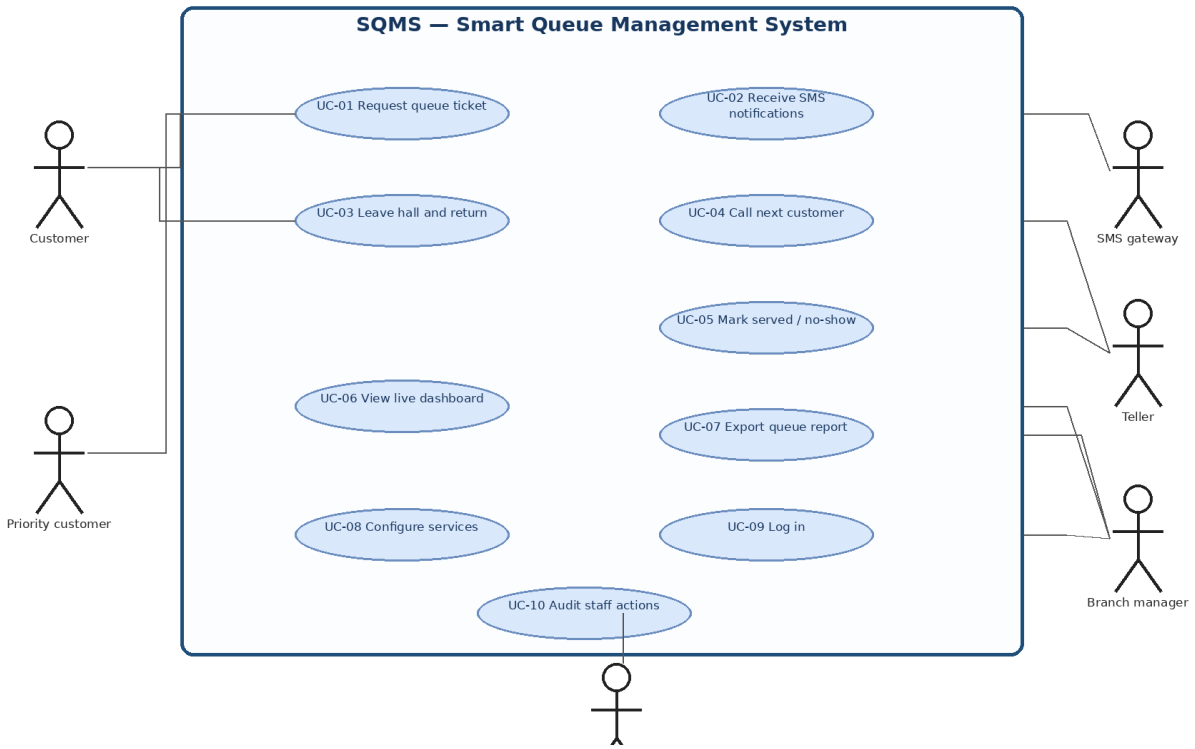
Staff screens work at 1366 × 768 on the latest two versions of Chrome and Edge; kiosk page works on a 10-inch Android tablet.

NFR-14 — Maintainability

Every API endpoint is documented; at least 5 automated tests run on every push (M5).

5. Use Case Model

The use case diagram identifies the external actors and the principal system services. The editable Draw.io source and PNG export are included in this submission package.



Actors: Customer, Priority customer, SMS gateway, Teller, Branch manager and Administrator. Use cases: request ticket; receive SMS; leave and return; call next; mark served/no-show; view dashboard; export report; configure services; log in; and audit staff actions.

5. Detailed Use Cases

UC-01 Request queue ticket

Actors: Customer; Priority customer; SMS gateway (secondary)
Precondition: Kiosk is running; branch is open; at least one service is configured.

Main flow

1. Customer selects a language and service.

2. Customer enters a mobile phone number.

3. System displays consent; customer accepts.

4. System issues the next sequential ticket and calculates estimated wait.

5. System displays ticket number, position and wait.

6. System sends confirmation SMS.

7. Customer may leave and return when alerted.

Alternative flows

1. Priority customer selects a priority category.

2. Invalid phone number: system shows an error in the selected language.

3. Consent refused: no ticket is issued; customer is directed to the floor guide.

4. SMS fails: ticket remains valid; failure is logged and retry is attempted.

Postcondition: Ticket exists with Waiting status and creation timestamp.

UC-04 Call next customer

Actors: Teller; SMS gateway (secondary)
Precondition: Teller is logged in and has opened a counter; at least one ticket is Waiting.

Main flow

1. Teller presses Call next.

2. System selects the first priority ticket, or the oldest standard ticket if no priority ticket waits.

3. System sets status to Called, records time, updates display and sends counter SMS.

4. Customer reaches counter; teller starts service.

5. Teller marks Served; completion time and service statistics update.

Alternative flows

1. No-show: after grace period teller marks No-show; ticket is skipped and customer may rejoin once.

2. Recall: teller recalls a Called ticket and notification/display are repeated.

3. Empty queue: system shows 'No customers waiting'.

Postcondition: Ticket is Served or No-show; queue statistics are updated.

6. User Stories and Acceptance Criteria

Story points use a Fibonacci-style scale. Priorities use MoSCoW. The board should contain all 14 stories and the acceptance criteria below.

US-01 — As a customer, I want to request a queue ticket at the branch by choosing a service and entering my phone number so that I do not need an unreliable paper ticket and my place is recorded.

Priority: Must | Points: 5 | Trace: PP1, PP3; FR-01, FR-02, FR-03

Given I am at the kiosk and have accepted the consent notice
When I choose a service and enter a valid phone number
Then I see my queue number, position and estimated wait within 3 seconds, and the number is unique and sequential for that service and day.

US-02 — As a customer, I want to receive an SMS with my queue number and estimated wait so that I can plan my time without standing in the hall.

Priority: Must | Points: 3 | Trace: PP1; FR-04, FR-05

Given I have a valid ticket and selected a language
When the ticket is issued
Then I receive an SMS within 10 seconds containing my ticket number, service and estimated wait in the selected language.

US-03 — As a customer, I want to receive an SMS when my turn is near so that I can leave the hall and return shortly before I am called.

Priority: Must | Points: 3 | Trace: PP1; FR-06, FR-11

Given I have a waiting ticket and three customers remain ahead
When the configured turn-near threshold is reached
Then I receive a message asking me to return, and my queue place remains valid while I am away.

US-04 — As a priority customer, I want to request a priority ticket without having to ask staff so that I am served earlier in accordance with the institution's policy and avoid standing for a long time.

Priority: Must | Points: 3 | Trace: PP2; FR-08

Given I am eligible for priority service
When I select the applicable priority category and submit the ticket request
Then my ticket is marked Priority and ordered according to the institution's approved priority policy.

US-05 — As a teller, I want to call the next customer with one action so that I serve in a clear, fair order and avoid repeated questions about who is next.

Priority: Must | Points: 5 | Trace: PP3, PP4; FR-09, FR-10

Given I am logged in and customers are waiting
When I press Call next
Then the first eligible priority ticket, or otherwise the oldest standard ticket, becomes Called and appears on the teller view and hall display.

US-06 — As a teller, I want to mark a customer as served or no-show and recall a called ticket so that The queue keeps moving and disputes are reduced.

Priority: Must | Points: 3 | Trace: PP3, PP4; FR-10, FR-11

Given a ticket is Called
When I mark it Served, Recall it, or mark it No-show after the 5-minute grace period
Then the ticket status and timestamps are updated, the queue is recalculated, and a no-show customer may rejoin once.

US-07 — As a branch manager, I want to see the live queue and today's waiting times so that I can react when the hall becomes crowded.

Priority: Must | Points: 5 | Trace: PP5; FR-12, FR-13

Given I am logged in as a manager
When I open the dashboard
Then I see customers waiting, average wait, customers served and busiest hours, refreshed within 2 seconds of a queue change.

6. User Stories and Acceptance Criteria (continued)

US-08 — As a staff member, I want to log in with a role so that Only authorised people can call tickets or view reports.

Priority: Must | Points: 3 | Trace: PP4, PP5; FR-16

AC-1 — Given I am a teller
When I try to open the manager dashboard
Then access is denied and the system does not disclose manager-only information.

AC-2 — Given I enter an incorrect password three consecutive times
When the third failed attempt is recorded
Then my account is locked for 5 minutes and the event is logged.

US-09 — As a branch manager, I want to export queue statistics by day and hour so that I can plan staffing for peak periods.

Priority: Should | Points: 3 | Trace: PP5; FR-14

Given queue data exists for the selected period
When I choose Export CSV
Then I receive a CSV containing ticket volume, average wait and customers served grouped by day and hour.

US-10 — As a customer, I want to use the kiosk and SMS in Kinyarwanda or English so that I understand every customer-facing message.

Priority: Should | Points: 5 | Trace: PP1, PP2; FR-15

Given I choose Kinyarwanda or English on the first screen
When I complete ticket registration and receive notifications
Then the kiosk and SMS content use the selected language consistently.

US-11 — As a customer without a phone, I want to get a ticket from the floor guide so that I am not excluded from the queue system.

Priority: Should | Points: 3 | Trace: PP2; FR-18

Given I do not have a mobile phone
When the floor guide or teller issues a ticket without a phone number
Then the system creates the ticket and shows its number on the hall display when it is near the front.

US-12 — As a administrator, I want to configure services, counters and the turn-near threshold so that The system fits the branch's operating policy.

Priority: Should | Points: 3 | Trace: PP2, PP5; FR-17

Given I am logged in as an administrator
When I change the turn-near threshold from 3 to 5
Then subsequent alerts are triggered when five customers remain ahead, and the change is audit-logged.

US-13 — As a customer, I want to reply to an SMS keyword to learn my ticket status so that I can check my position without returning to the branch.

Priority: Could | Points: 3 | Trace: PP1; FR-20

Given I hold a Waiting ticket and the SMS status service is available
When I reply with a recognised status keyword in English, Kinyarwanda, French or Swahili
Then I receive my current ticket status, queue position and estimated wait in the corresponding language.

US-14 — As a customer in the hall, I want to see current numbers on a display screen so that I can follow service progress visually.

Priority: Could | Points: 2 | Trace: PP3; FR-19

Given a counter calls a ticket
When the queue event is processed

Then the hall display shows the ticket number and counter within 2 seconds.

7. MoSCoW Prioritised Backlog

Priority	Stories	Total points
Must — core Sessions 4 and 5	US-01 to US-08	30
Should	US-09, US-10, US-11, US-12	14
Could	US-13, US-14	5
Won't (this course)	Core-banking integration, appointment booking, native mobile app, payments, multi-branch analytics	—
TOTAL ESTIMATED STORY POINTS	14 stories	49

Must scope: US-01 to US-08 (30 points). Should scope: US-09 to US-12 (14 points). Could scope: US-13 to US-14 (5 points). Total: 49 points. US-10 is 5 points in this revision.

8. Traceability Matrix

Pain point	Functional requirements	NFRs	Stories
PP1 Long/unpredictable waiting; cannot leave queue	FR-01, 04, 05, 06, 07, 11, 15, 20	NFR-01, 02, 09, 10	US-01, 02, 03, 10, 13
PP2 No seating/priority for vulnerable customers	FR-08, 15, 17, 18	NFR-07, 08	US-04, 10, 11, 12
PP3 Unclear order and disputes	FR-01, 03, 09, 10, 19	NFR-03, 12	US-01, 05, 06, 14
PP4 Staff workload and interruptions	FR-09, 10, 16, 21	NFR-03, 06	US-05, 06, 08
PP5 No queue data for management	FR-12, 13, 14, 16, 17, 21	NFR-04, 05, 11	US-07, 08, 09, 12
Privacy/legal	FR-02, 21	NFR-05, 06	US-01, 08

9. User Validation Record

The team reports that validation was completed with two anonymised participants (V1 and V2), using five questions and feedback against the Must stories. The recovered source SRS contained blank validation fields, and the exact interview notes were not available in the materials retrieved for this reconstruction. The table below therefore records the validation framework and the changes explicitly supplied by the team; it does not invent quotations or claim verbatim responses.

Validation participants and questions

ID	Participant	Status
V1	Anonymised user representative; role to be copied from signed team notes.	Team reports completed
V2	Anonymised staff representative; role to be copied from signed team notes.	Team reports completed

Five validation questions to retain in the record

1. Is requesting a ticket using a service and phone number clear and understandable?
2. Does the queue number, position and estimated wait provide the information you need?
3. Are the turn-near and called-to-counter notifications useful and understandable?
4. Does the priority-ticket process feel fair, respectful and consistent with the branch policy?
5. Can staff easily call the next customer, manage no-shows and understand the dashboard?

Must-story feedback register

Story	Feedback record	Change incorporated
US-01	Use the V1/V2 signed notes to insert the actual per-story feedback before submission; no verbatim comment is fabricated in this reconstruction.	No specific wording change was supplied in the recovered revision notes; verify against the team's validation log.
US-02	Use the V1/V2 signed notes to insert the actual per-story feedback before submission; no verbatim comment is fabricated in this reconstruction.	No specific wording change was supplied in the recovered revision notes; verify against the team's validation log.
US-03	Use the V1/V2 signed notes to insert the actual per-story feedback before submission; no verbatim comment is fabricated in this reconstruction.	No specific wording change was supplied in the recovered revision notes; verify against the team's validation log.
US-04	Use the V1/V2 signed notes to insert the actual per-story feedback before submission; no verbatim comment is fabricated in this reconstruction.	No specific wording change was supplied in the recovered revision notes; verify against the team's validation log.
US-05	Use the V1/V2 signed notes to insert the actual per-story feedback before submission; no verbatim comment is fabricated in this reconstruction.	No specific wording change was supplied in the recovered revision notes; verify against the team's validation log.
US-06	Use the V1/V2 signed notes to insert the actual per-story feedback before submission; no verbatim comment is fabricated in this reconstruction.	No specific wording change was supplied in the recovered revision notes; verify against the team's validation log.
US-07	Use the V1/V2 signed notes to insert the actual per-story feedback before submission; no verbatim comment is fabricated in this reconstruction.	No specific wording change was supplied in the recovered revision notes; verify against the team's validation log.
US-08	Use the V1/V2 signed notes to insert the actual per-story feedback before submission; no verbatim comment is fabricated in this reconstruction.	Acceptance criteria separated into AC-1 (role access) and AC-2 (three failed passwords lock account for 5 minutes).

Additional recorded changes: US-10 increased from 2 to 5 points; FR-20 status keywords extended to English, Kinyarwanda, French and Swahili; FR-21 added (exact approved wording must be checked); MoSCoW total corrected to 49 points.

10. Risks, Dependencies and Acceptance

Risks and mitigations

Risk	Mitigation / treatment
SMS gateway unavailable or delayed	Keep ticket valid if SMS fails; log failure; retry safely; display ticket number on kiosk.
Concurrent ticket requests create duplicate numbers	Use a database transaction/locking strategy and enforce a unique constraint for branch, service, date and number.
Personal data is exposed	Collect consent, mask phone numbers for staff, enforce role access, use TLS and apply retention rules; obtain legal review.
Priority policy differs by institution	Make the policy configurable and validate it with the branch before deployment.
Low time/velocity for Must scope	Protect Must stories; reduce non-essential detail in dashboard before dropping core queue functionality.

Acceptance conditions for M2

The requirements baseline is acceptable when each functional requirement is uniquely identified and testable; NFRs have measurable criteria; each user story has Given/When/Then acceptance criteria; priorities and points sum correctly; traceability links M1 pain points to requirements and stories; the editable use case diagram is included; and validation notes are reviewed and signed off by the team.

Milestone 2 deliverables checklist

Deliverable	Location	Status
SRS PDF	This document	Prepared
Use case diagram source and PNG	T6-M2-UseCase.drawio / .png	Included
14 stories with acceptance criteria	Jira/Trello board + copy-paste file	Copy-paste file included; board screenshot/link must be added by team
MoSCoW and traceability	Sections 7–8	Included
V1/V2 validation notes	Section 9 and team notes	Framework included; insert exact signed feedback
sources.md and ai-use-log.md	docs/	Included in package
GitHub tag/release m2	Repository	Team action required

Appendix A — Glossary and Data Handling

SQMS: Smart Queue Management System. ETW: estimated time/wait. SMS: Short Message Service. REST: Representational State Transfer. CSV: Comma-Separated Values. TLS: Transport Layer Security. WCAG: Web Content Accessibility Guidelines.

Data handling principles

Phone numbers are collected only for queue-related communication after clear consent. The system should display masked numbers to staff, limit access by role, protect traffic using TLS, store passwords using a suitable password hash, expire inactive sessions and delete or anonymise phone numbers after the defined retention period. The proposed retention period in NFR-05 is 30 days and requires legal/operational confirmation before production use.

Queue status model

Waiting → Called → Serving → Served. A Called ticket may be recalled or marked No-show after the grace period. A no-show customer may rejoin once according to FR-11. All transitions should be timestamped; staff-initiated changes should be auditable under FR-21.

Concurrency and integrity

Ticket generation must be atomic for each branch, service and day. The implementation should use a transaction and a database-level uniqueness guarantee so two simultaneous requests cannot receive the same number. A suitable PostgreSQL locking strategy should be selected and tested under concurrent requests.

Appendix B — References

[R1] International Organization for Standardization, International Electrotechnical Commission, & Institute of Electrical and Electronics Engineers. (2018). ISO/IEC/IEEE 29148:2018 Systems and software engineering — Life cycle processes — Requirements engineering. ISO. https://www.iso.org/standard/72089.html

[R2] Republic of Rwanda. (2021). Law No. 058/2021 of 13/10/2021 relating to the protection of personal data and privacy. Official Gazette of the Republic of Rwanda, Special issue of 15/10/2021. https://www.risa.gov.rw/fileadmin/user_upload/RISA/Publications/2.Laws/Law_relating_to_the_protection_of_personal_data_and_privacy.pdf

[R3] World Wide Web Consortium. (2023). Web Content Accessibility Guidelines (WCAG) 2.2. https://www.w3.org/TR/WCAG22/

M1 discovery sources

[S1] Rwanda Today. (2022, April 3). Roll out online and mobile services, members tell SACCOs. https://rwandatoday.africa/rwanda/business/roll-out-online-and-mobile-services-members-tell-saccos-3770450

[S2] The New Times. (2019, March 29). Govt searches for firm to automate SACCO services. https://www.newtimes.co.rw/article/165302/News/govt-searches-for-firm-to-automate-sacco-services/amp

[S3] Icyemezo, H., Mung'atu, J. K., & Ndanguza, D. (2017). Multi-channel queuing modeling on system delivery service: A case study of Banque Populaire du Rwanda Kimironko branch. International Journal of Science, Engineering and Technology. https://www.ischolar.in/index.php/IJSET/article/view/160911

[S4] National Bank of Rwanda. (n.d.). Mystery shopping report: Service delivery in financial service providers. https://www.bnr.rw/documents/MYSTERY_SHOPPING_REPORT.pdf

[S5] The New Times. (n.d.). Mad rush to reactivate dormant accounts as SACCOs get automated. https://www.newtimes.co.rw/article/7445/news/rwanda/mad-rush-to-reactivate-dormant-accounts-as-saccos-get-automated/amp

[S6] Wavetec. (n.d.). Queue management solutions for banks. https://www.wavetec.com/

[S7] Qmatic. (n.d.). Qmatic Experience Cloud. https://www.qmatic.com/

[S8] SourceForge. (n.d.). Qmatic Experience Cloud vs Wavetec Queue Management. https://www.sourceforge.net/software/compare/Qmatic-Experience-Cloud-vs-Wavetec-Queue-Management

[S9] Qminder. (n.d.). Top 5 Qless alternatives for a queuing system. https://www.qminder.com/blog/queue-management/top-5-qless-alternatives-for-a-queuing-system/

Appendix C — Submission Notes

Before final upload, the team must check the exact V1/V2 validation notes and approved FR-21 wording against its working records. Add the Jira/Trello board URL and screenshot. Confirm that all team members agree to the final scope and that the repository is ready for the m2 release tag.

Suggested release title: Milestone 2 — Requirements Engineering and System Analysis. Suggested tag: m2. Do not create a duplicate tag if m2 already exists; inspect the repository first.

Repository: https://github.com/namahorojanvier1986-blip/secd-2026-team6-sqms

End of document.